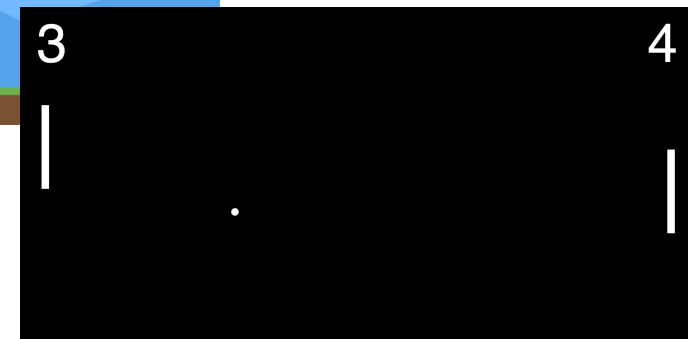
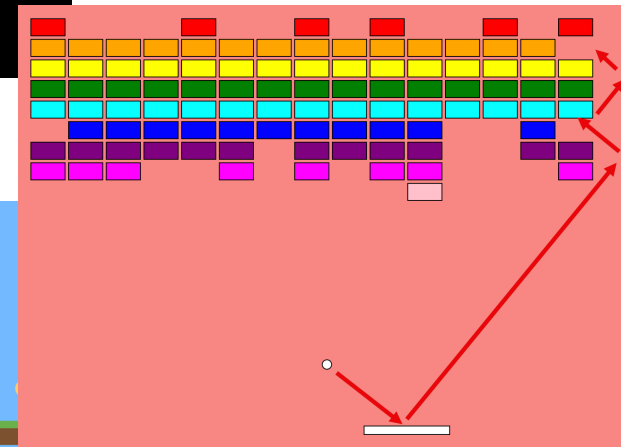
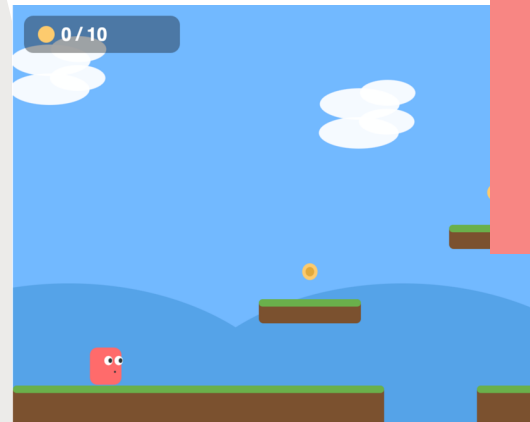
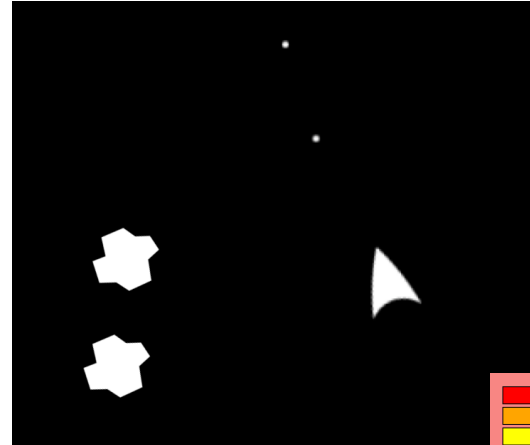


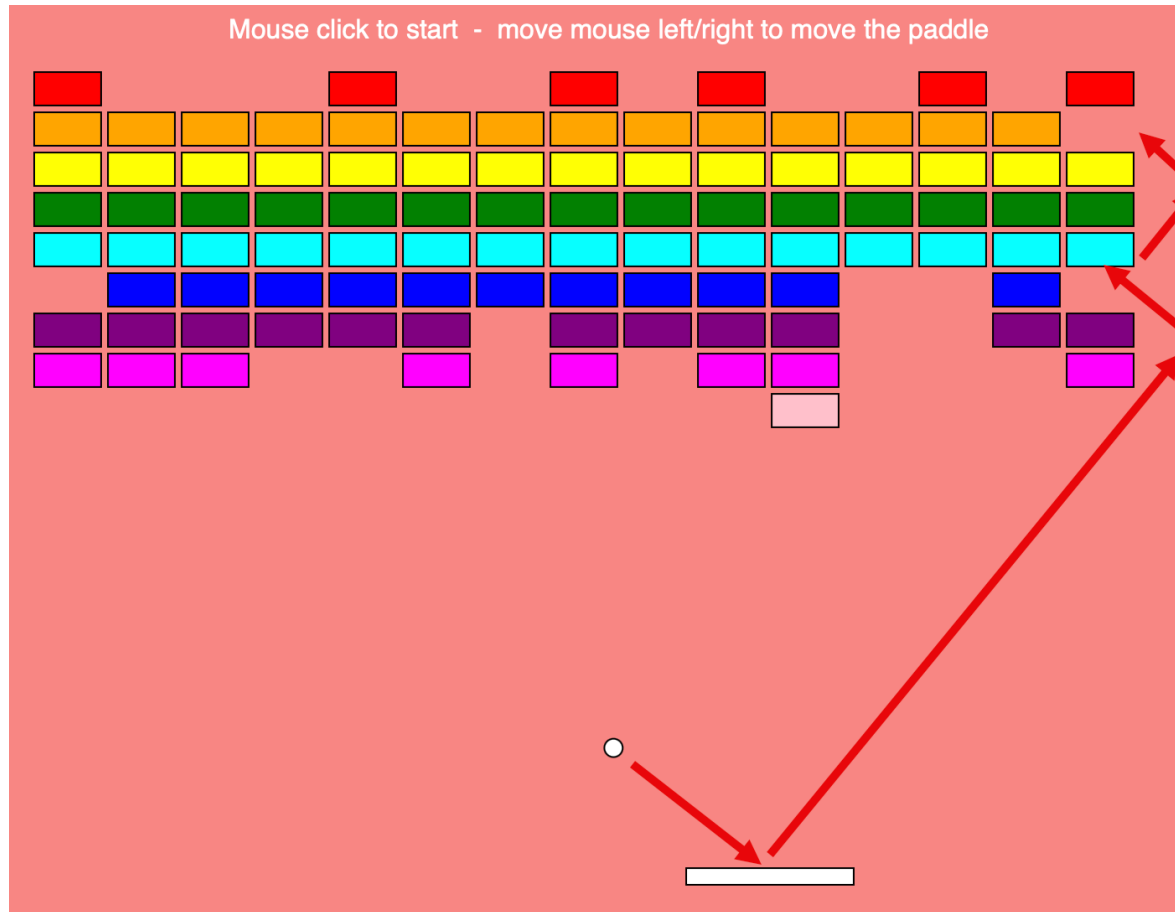
Informatics Summer Camp

Coding and Web
Development

p6play JavaScript
web games



Take a look at the game "Breakout"



- ✓ ____SummerCamp2026
 - > challenge-01
 - > challenge-02
 - > color_reference
 - ✓ games
 - > 2D_platformer
 - > asteroids
 - ✓ breakout
 - index.html
 - script.js
 - > flappy_bird
 - > pong
 - > images
 - > lesson-01

Ingredients of a game (examples from Breakout game)

1. A rectangle (canvas) to draw in

- Breakout: pink background



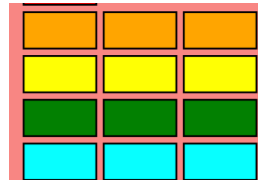
2. Player-controlled object (sprite)

- Breakout: white rectangle
- + logic (code!) to detect user inputs (mouse and keyboard)



3. Other objects (sprites)

- Breakout: moving white ball
- Breakout: colored blocks
- + logic for collisions
 - Ball: bounce
 - Colored block: delete



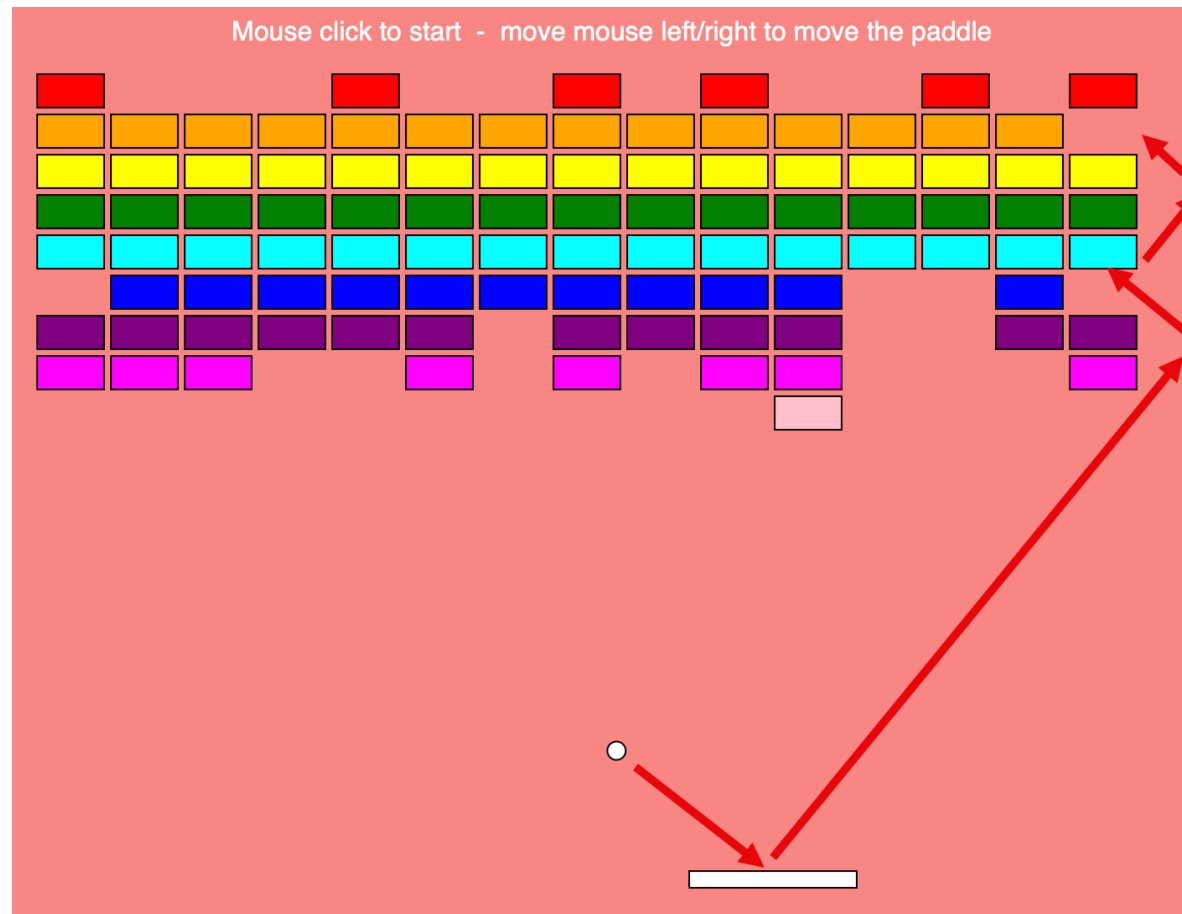
4. Text: Instructions / score / timer

- Breakout: instructions

Mouse click to start

5. Sound

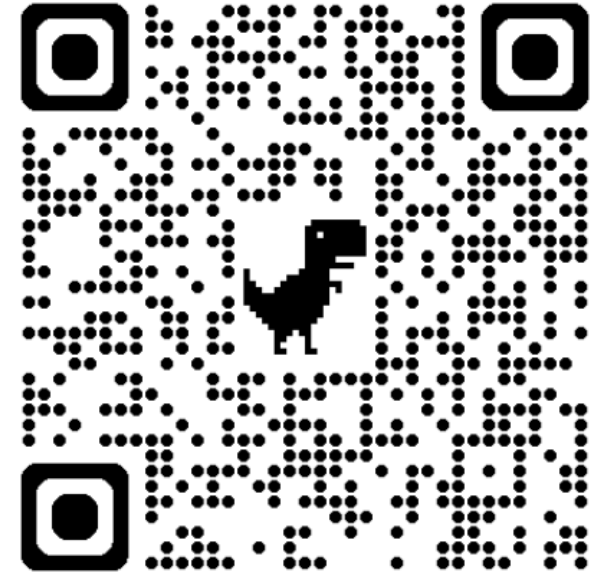
- Breakout: when block deleted



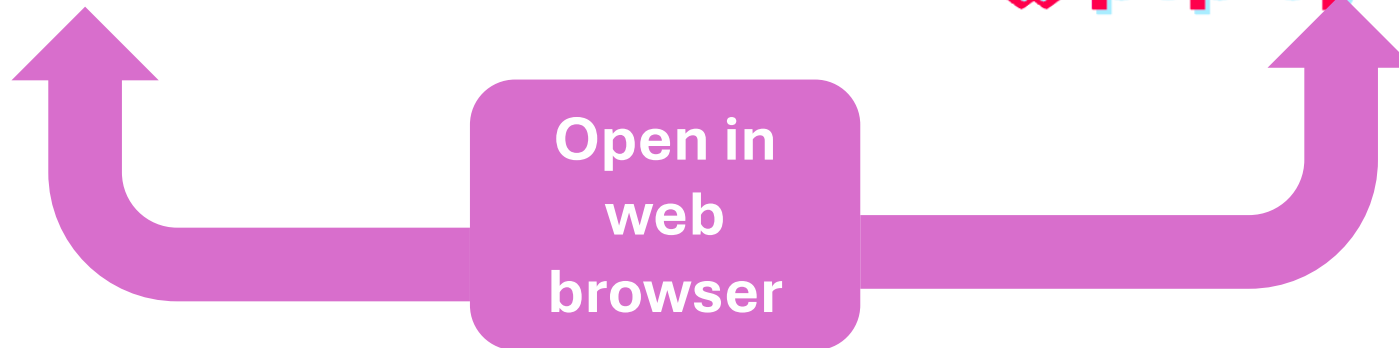
Lesson Resources



<https://github.com/dr-matt-smith/SummerCamp2026>



<https://p5play.org/learn/sprite.html>



If 2D barcodes not working for you ...

- In web browser search for
github
dr-matt-smith
- choose project
SummerCamp2026

dr-matt-smith

Overview Repositories 1.2k Projects Packages Stars 3

SummerCamp2026 Public
Forked from stephensheridan/SummerCamp2024
Informatics Summer Camp resources
JavaScript

2026-FDEev-Front-End-Dev-start-he Public
FED public lab resources for my TU Dublin student

php-crash-course Public
Code listings and exercise solutions from my book: PHP Crash Course (No Starch Press)
PHP 23 11

PacktPublishing/Unity-2023-Cookbook Public
Unity 2023 Cookbook
PHP 48 13

pandoc-thesis-template Public
Uses Github actions to create high quality PDF thesis from simple text MarkDown files
TeX 16 23

pdo-crud-for-free-repositories Public
A PHP library using PDO and reflection to minimise communication with databases. The library uses ORM - which map between DB tables and corresponding PHP classes
PHP 5

Matt Smith
dr-matt-smith · he/him
computing academic & open-source software developer
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166 followers · 2 following

Ireland
matt.smith@tudublin.ie

999 contributions in the last year
Contribution settings

Jun Jul Aug Sep Oct Nov Dec Jan Feb Mar
Mon No contributions on June

download
project
ZIP file

The screenshot shows the GitHub interface for the repository 'dr-matt-smith / SummerCamp2026'. The repository is public and forked from 'stephensheridan/SummerCamp2024'. The 'Code' menu is open, showing options for cloning and downloading. The 'Download ZIP' option is highlighted with a red circle. The 'Code' button is also circled in red. The repository has 1 branch and 0 tags. The commit history shows 'dr-matt-smith' updated notes. The file list includes '_old_p5', 'games', and 'images'.

dr-matt-smith / SummerCamp2026

Type / to search

Code Pull requests Agents Actions Projects Wiki Security and quality

SummerCamp2026 Public

forked from [stephensheridan/SummerCamp2024](#)

main 1 Branch 0 Tags

This branch is 3 commits ahead of stephensheridan/SummerCamp2024

Contribute Sync fork

dr-matt-smith updated notes

_old_p5

games

images

1: Click "Code" menu button

2: Choose Download ZIP

Lesson 1: Setup and Draw

- Work in project **SummerCamp2026**
- Start working in folder **lesson-01**
- Using a text editor, look at the 2 files in folder **lesson-01**



Lesson 1: Setup and Draw

- Look at **lesson-01** files:

index.html

```
1 <!DOCTYPE html>
2 <html lang="en">
3   <head>
4     <meta charset="utf-8">
5     <title>a p5play sketch</title>
6     <!-- q5 and p5play Javascript libraries -->
7     <script src="https://cdn.jsdelivr.net/npm/q5@4.7.3/q5.min.js"></script>
8     <script src="https://p5play.org/v3/planck.min.js"></script>
9     <script src="https://p5play.org/v3/p5play.js"></script>
10    <!-- Your script -->
11    <script src="script.js"></script>
12  </head>
13  <body>
14    <main>
15    </main>
16  </body>
17 </html>
```

script.js

```
function setup() {
  createCanvas(400, 400);
}

function draw() {
  background('red');
}
```

Lesson 1: Setup and Draw

- In a browser, open **index.html** file from folder **lesson-02**
- Should see a blank red canvas.
- **NOTE:** when you make a change to your code, refresh the browser window to see the results



Challenge 1: Setup and Draw

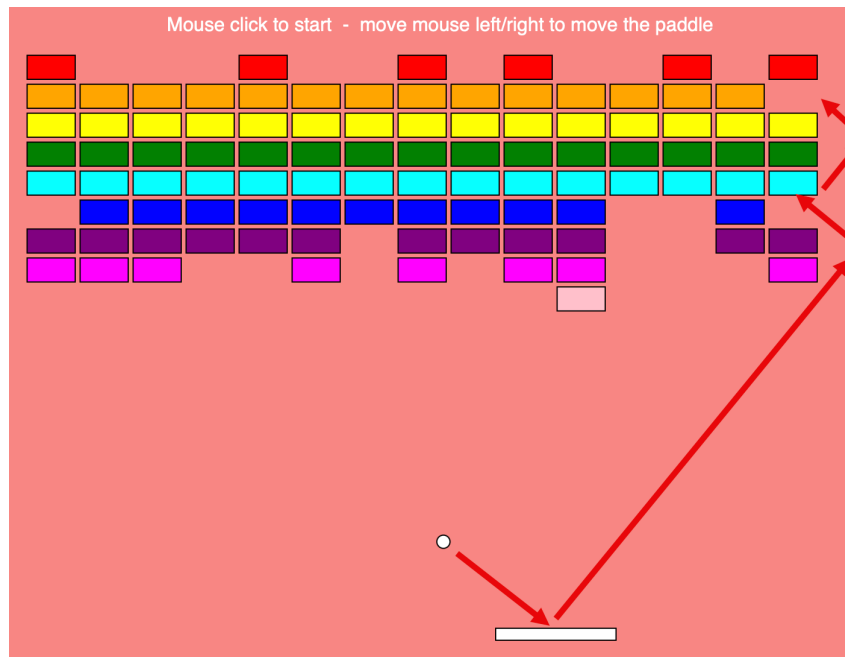
Lesson challenge 1:

Change the canvas background color to: **grey**



Game challenge 1:

Try to change to a color like the **Breakout** game background



	ivory #FFFFFF0	khaki #F0E68C	lavender #E6E6FA
	lightblue #ADD8E6	lightcoral #F08080	lightcyan #E0FFFF
	lightpink #FFB6C1	lightsalmon #FFA07A	lightseagreen #20B2AA
	lightyellow #FFFFE0	lime #00FF00	limegreen #32CD32
	mediumaquamarine #66CDAA	mediumblue #0000CD	mediumorchid #BA55D3

HINT

Lesson 2: Sprites

- Work in folder: **lesson-02**
- We will only be editing file:
script.js
- (index.html never changes)



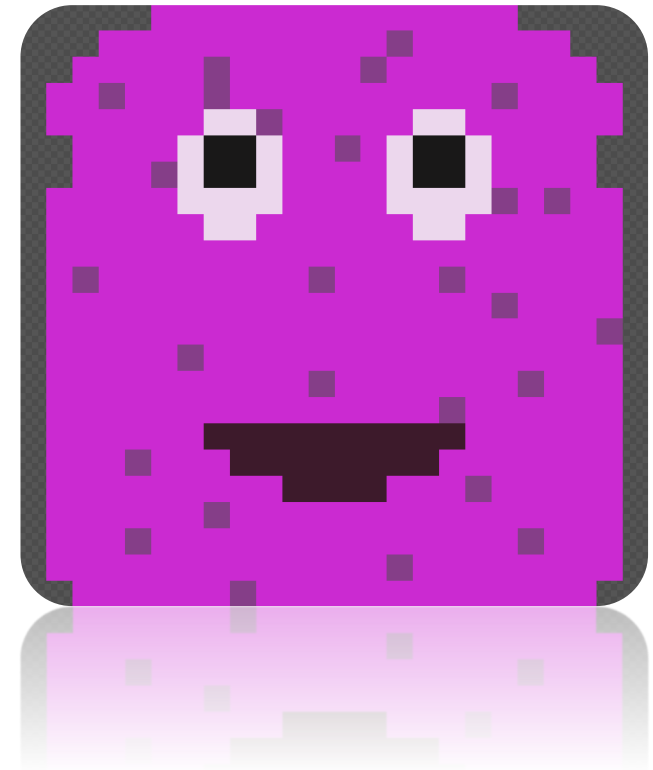
Lesson 2: Sprites

What is a sprite?

A sprite is a ghost!

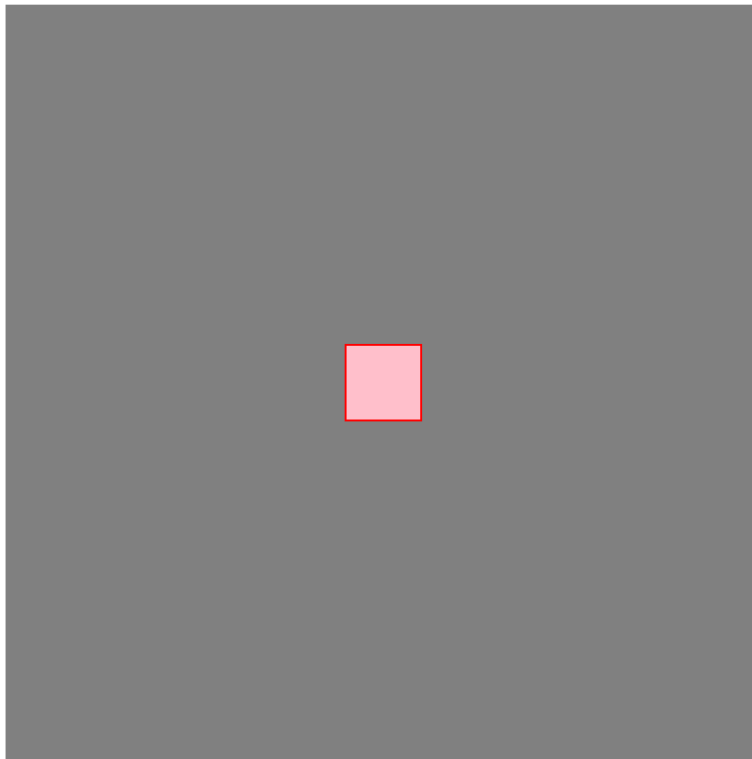
Video game developers use the word "sprite" to refer to characters, items, or anything else that moves above a background.

The `new Sprite()` constructor creates a sprite object, which contains variables that define a sprite's position, size, and appearance.



Lesson 2: Sprites

- Modify file: lesson-02/**script.js** to the following – and view in browser



```
1 let sprite1;
2
3 function setup() : void {
4   new Canvas(600, 400);
5   sprite1 = new Sprite()
6   sprite1.x = 200;
7   sprite1.y = 200;
8   sprite1.w = 40;
9   sprite1.h = 40;
10  sprite1.color = 'pink';
11  sprite1.stroke = 'red';
12 }
13
14 function draw() : void {
15   background('grey');
16 }
```

Lesson 2: Sprites

- We can pass values to **Sprite** (**x**, **y**, **w**, **h**)
- so less code to write

```
let sprite1;

function setup() : void {
  new Canvas(600, 400);

  sprite1 = new Sprite(200, 200, 40, 40)
  sprite1.color = 'pink';
  sprite1.stroke = 'red';
}

function draw() : void {
  background('grey');
}
```

Lesson 2: Sprites

- can set random location for sprite using function **random()**
- Some examples:

```
// Random number between 0 and 1  
var num = random(0,100);
```

```
// Random x position  
var x = random(width);
```

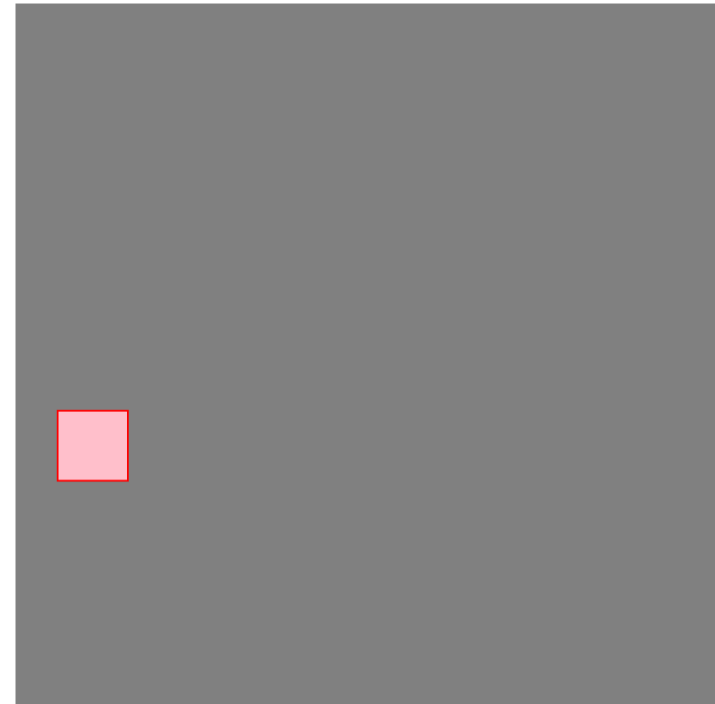
```
// Simulate a dice roll  
var roll = random(1,6);
```

```
// Random y position  
var y = random(height);
```

Lesson 2: Sprites

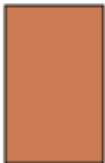
- Modify **lesson-02/ script.js**
- see what happens when refreshed each time in browser

```
1  let sprite1;  
2  
3  function setup() : void {  
4    new Canvas(600, 400);  
5  
6    let x = random(width);  
7    let y = random(height);  
8  
9    sprite1 = new Sprite(x, y, 40, 40)  
10   sprite1.color = 'pink';  
11   sprite1.stroke = 'red';  
12 }  
13  
14 function draw() : void {  
15   background('grey');  
16 }
```



Lesson 2: Sprites

- **Sprites** have many different properties. Experiment with the properties show below.

 x   <pre>sprite.x = 175;</pre>	 y   <pre>sprite.y = 30;</pre>	 width   <pre>sprite.w = 200;</pre>	 height   <pre>sprite.h = 80;</pre>
 color   <pre>sprite.color = 'pink'; sprite.stroke = 'red';</pre>	 rotation   <pre>sprite.rotation = 45;</pre>	 diameter   <pre>sprite.d = 40;</pre>	 textSize   <pre>sprite.textSize = 40; sprite.text = "p5";</pre>

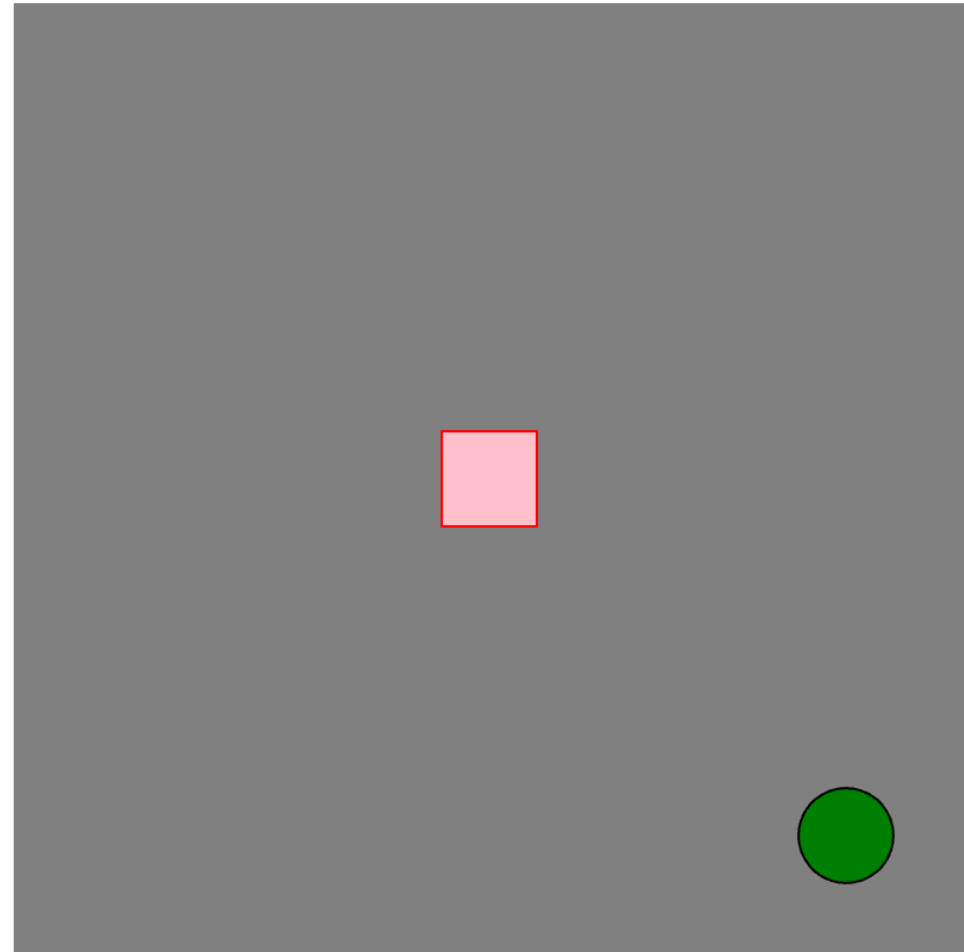
Challenge 2: Sprites

Lesson challenge 2: Create second **Sprite** called **ball**

- circular with diameter **20**
- positioned bottom right corner
- filled with green and with a black border

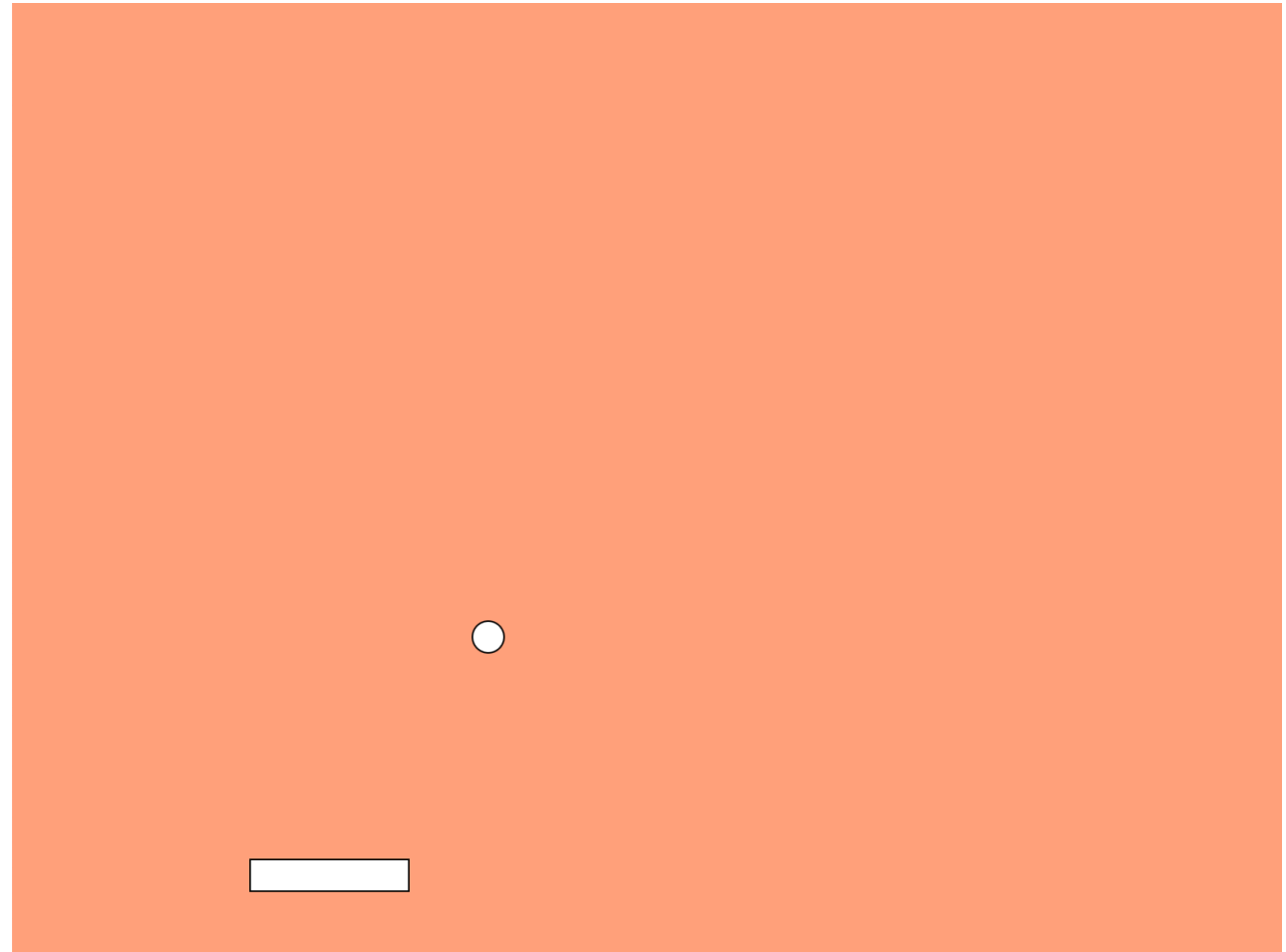
HINT – pass in (x, y, d) to create a circular Sprite

```
new Sprite(100, 200, 50);
```



Challenge 2: Sprites

- **Game challenge 2:** Create **Sprites** for Breakout
 - **ball:** a white circle
 - **paddle:** a white rectangle



QUESTION

How do we add text to a Sprite?

Raise your hand if you know the answer.

Lesson 3: Sprite Movement

- Work in folder: **lesson-03**



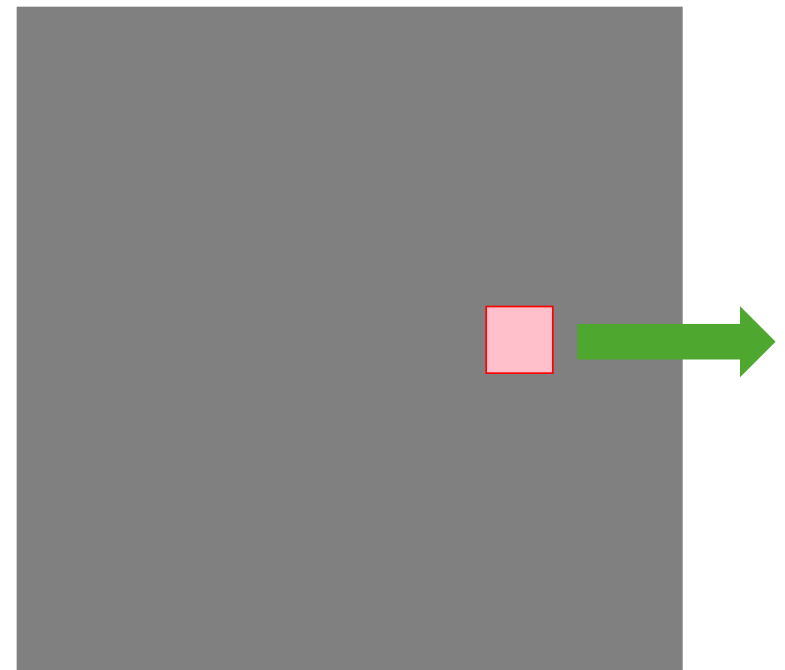
```
> lesson-01
> lesson-02
v lesson-03
  <> index.html
  JS script.js
```

Lesson 3: Sprite Movement

- Can move Sprites by giving them **x** or **y velocity**



```
1 let sprite1;  
2  
3 function setup() : void {  
4   new Canvas(600, 400);  
5  
6   sprite1 = new Sprite(200, 200, 40, 40);  
7   sprite1.color = 'pink';  
8   sprite1.stroke = 'red';  
9  
10  sprite1.vel.x = 1;  
11 }  
12  
13 function draw() : void {  
14   background('grey');  
15 }
```



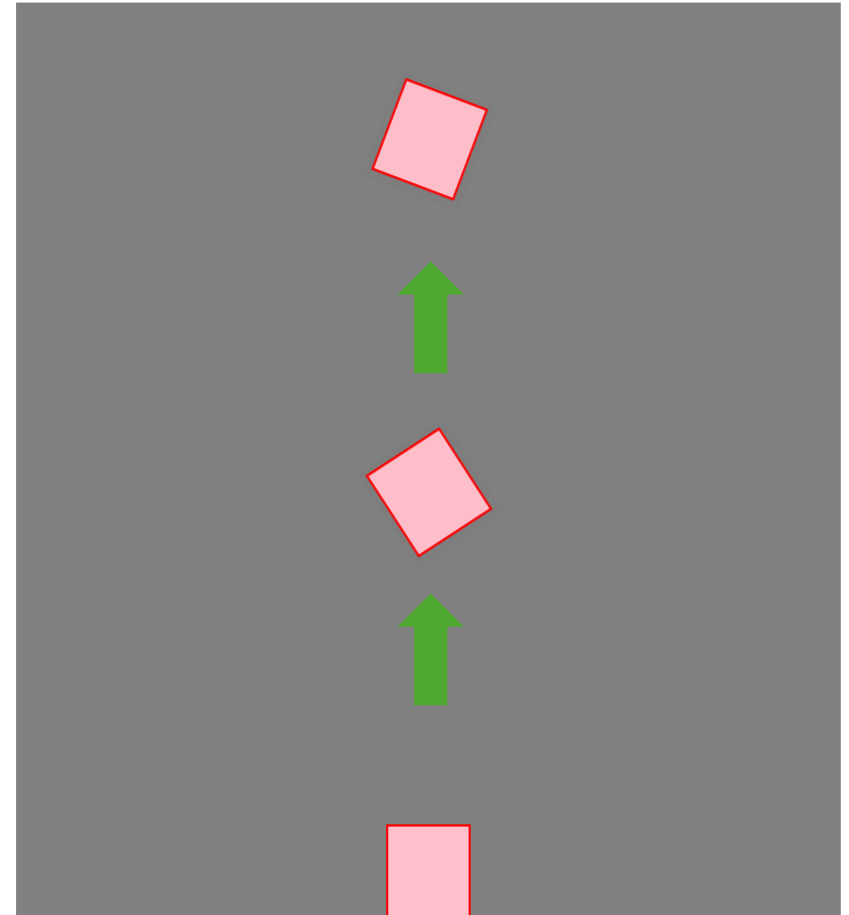
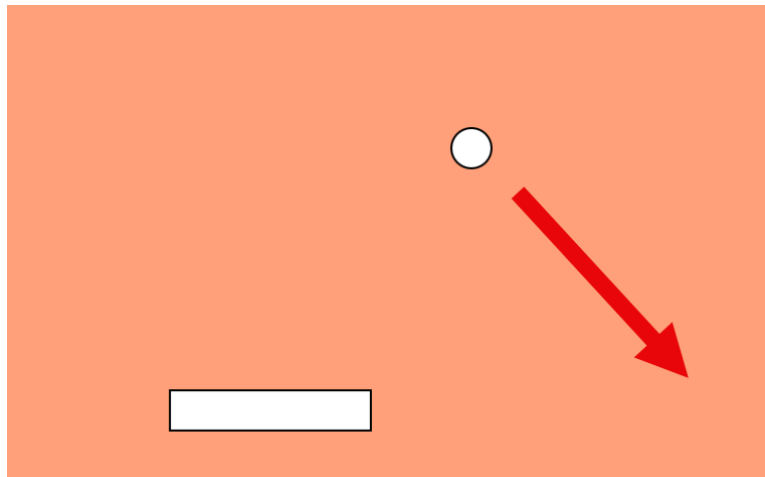
Challenge 3: Sprite Movement

Lesson challenge 3:

- Make Sprite move upwards
- & give the Sprite some rotation
 - using the **rotationSpeed** property.

Game challenge 3:

- Make white ball move down and to the right



QUESTION

Name the property that controls the colour of a Sprite's border?

Raise your hand if you know the answer.

Lesson 4: Where's my Sprite?

- Work in folder: **lesson-04**



```
> lesson-01
> lesson-02
> lesson-03
v lesson-04
    <> index.html
    JS script.js
```

Lesson 4: Where's my Sprite?

- PROBLEM: We can no longer see Sprite when its **x** and **y** position goes outside the canvas area.
- SOLUTION: We can make sprite bounce right and left by using a **test** in the **draw** function.

```
10  sprite1.vel.x = 5;
11  }
12
13  function draw() : void {
14    background('grey');
15
16    if (sprite1.x > 400)
17      sprite1.vel.x = -5;
18
19    if (sprite1.x < 0)
20      sprite1.vel.x = 5;
21  }
```

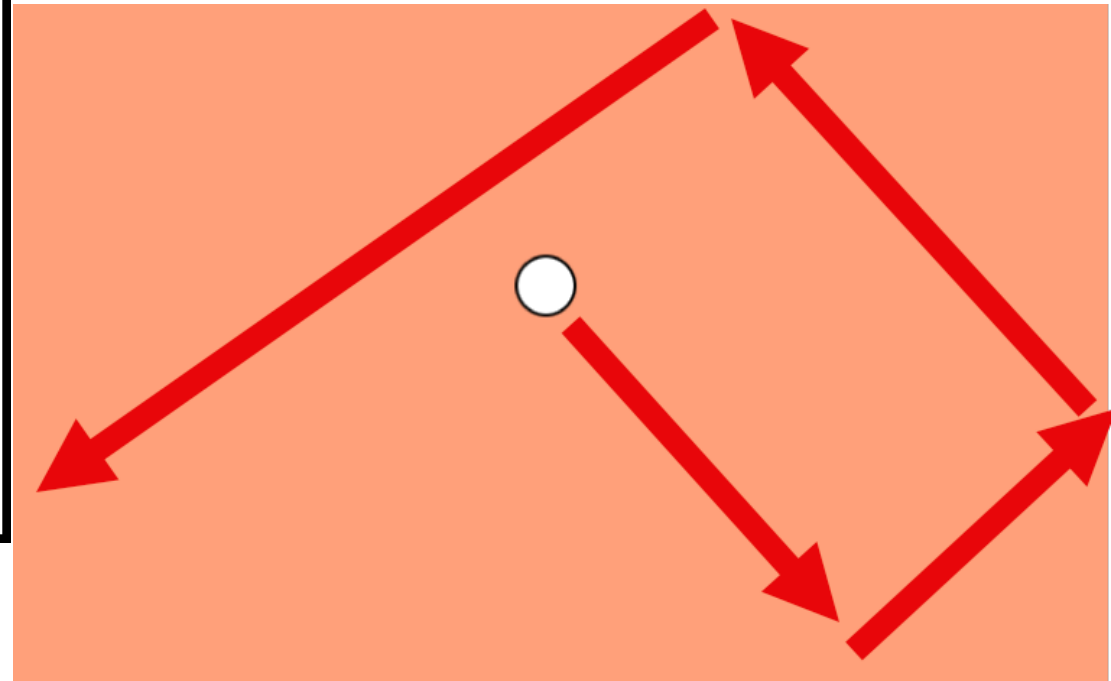


Change x
direction
to -1

Lesson 4: Where's my Sprite?

- TIP: for general purpose 'bounce' we reverse the velocity
 - So instead of 1 or -1, we set new velocity to **(-1 * old velocity)**
 - This makes bouncing ball very easy to implement....

```
13 function draw() : void {  
14   background('grey');  
15  
16   if (sprite1.x > 400)  
17     |   sprite1.vel.x = -1 * sprite1.vel.x;  
18  
19   if (sprite1.x < 0)  
20     |   sprite1.vel.x = -1 * sprite1.vel.x;  
21 }  
    And the same for sprite.vel.y
```



Challenge 4: Where's my Sprite?

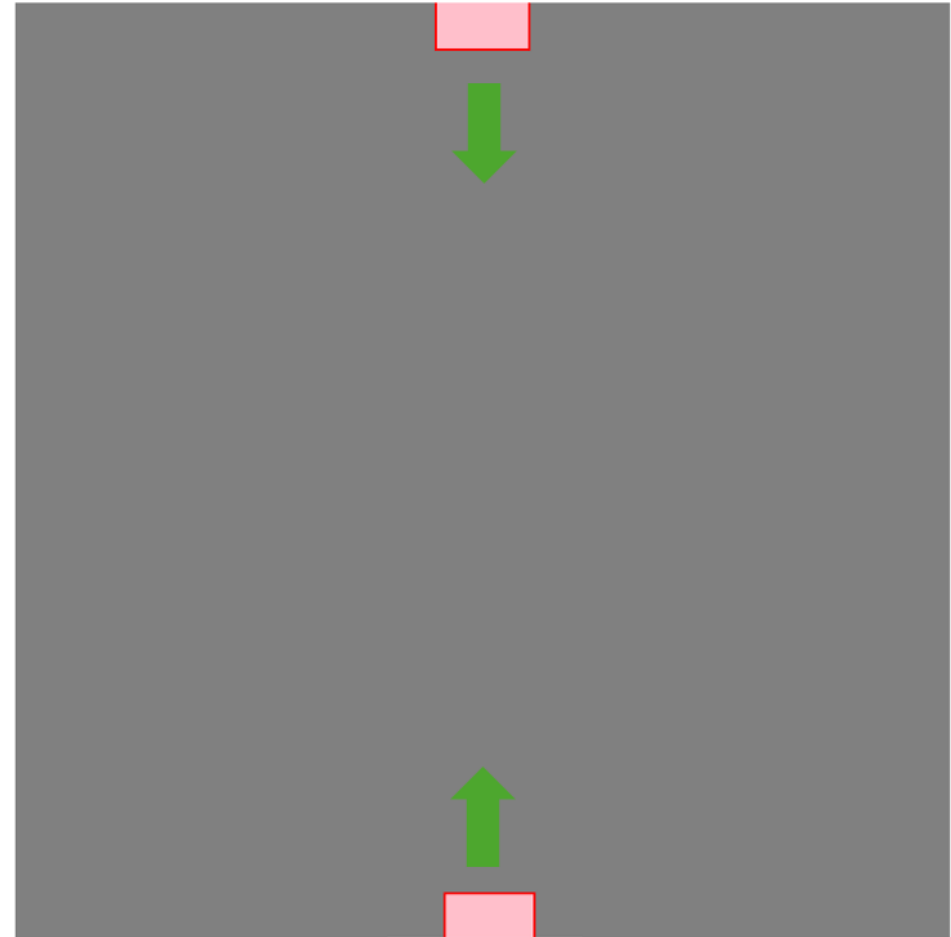
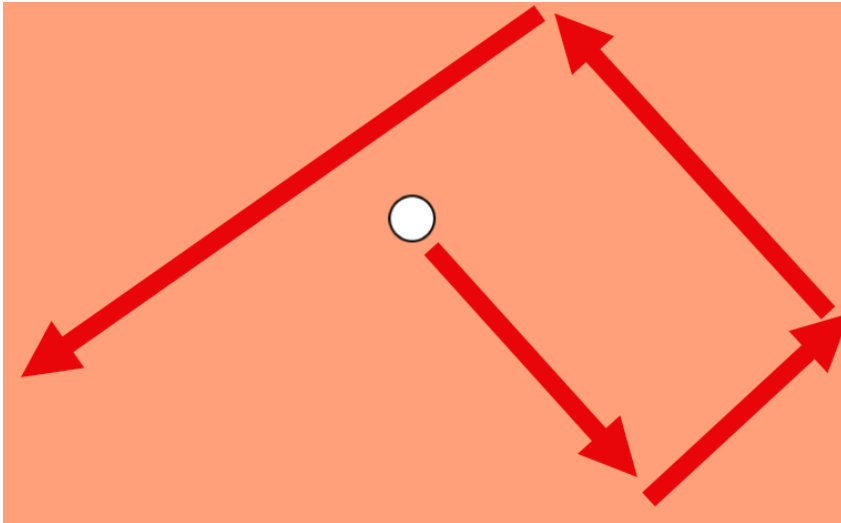
Lesson challenge 4:

Make the Sprite bounce up and down instead of right to left.

Game challenge 4:

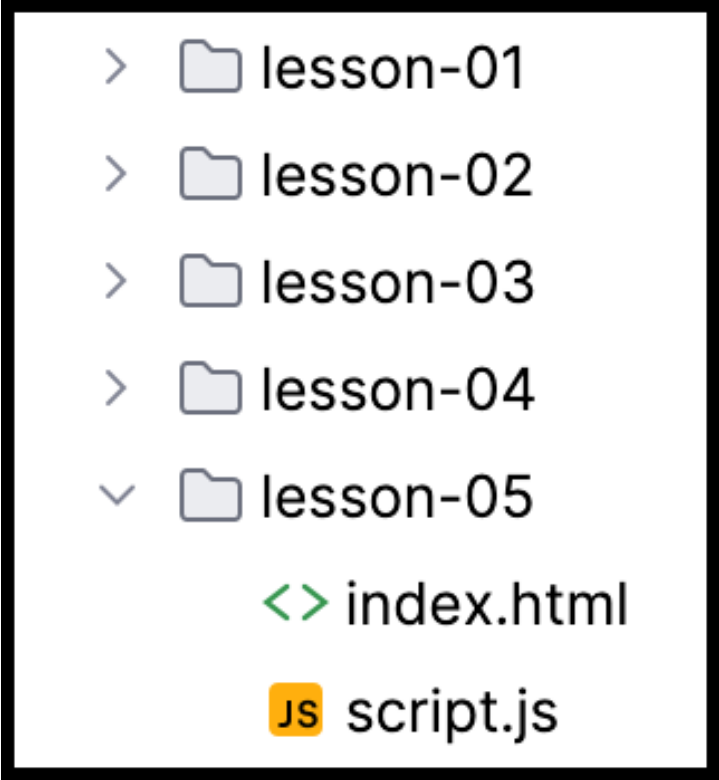
Make **ball** bounce off all 4 walls

And start it with both an 'x' and 'y' velocity



Lesson 5: Sprite Images

- Work in folder: **lesson-05**



```
> lesson-01
> lesson-02
> lesson-03
> lesson-04
v lesson-05
  <> index.html
  JS script.js
```

Lesson 5: Sprite Images

- Sprites not limited to basic shapes for Sprites
- can load **images** from web into our Sprites
 - or from inside project folder
- Can provide image to Sprite()
 - `new Sprite(image, x, y, w, h)`
- can use **preload()** function
 - to ensure all images that we need are loaded and ready to use before things get started
 - E.g.

```
let monsterImage;
```

```
function preload() {
```

```
    monsterImage = loadImage('images/monster.png');
```

```
}
```

Lesson 5: Sprite Images

- In these lessons we will use the following images
- In your lesson-05 folder, create a folder “**images**”
 - And copy these 5 images there from the main project “images” folder



tudublin.png



monster.png



moon.png



earth.png

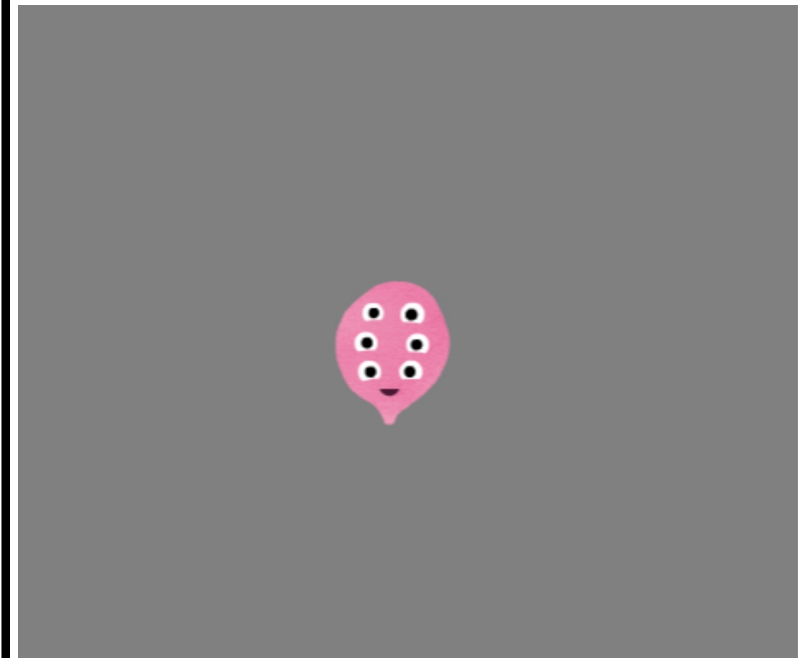


ship.png

```
▼ lesson-05
  ▼ images
    earth.png
    monster.png
    moon.png
    ship.png
    tu_dublin_logo.png
  <> index.html
  JS script.js
```

Lesson 5: Sprite Images

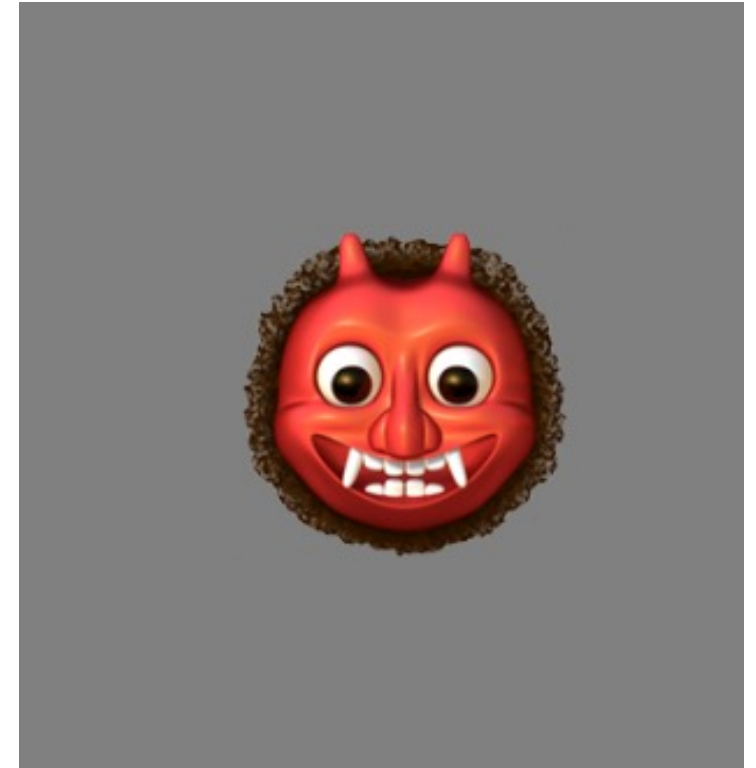
```
1 let monster, monsterImage; ←
2
3 function preload() : void {
4   monsterImage = loadImage('images/monster.png'); ←
5 }
6
7 function setup() : void {
8   new Canvas(400, 400);
9   monster = new Sprite(monsterImage, 200, 200, 72, 69) ↑
10 }
11
12 function draw() : void {
13   background('grey');
14 }
```



Lesson 5: Sprite Images: emojis

- Bonus – can use **emojis** instead of images!
 - Set using `.image()` property

```
let monster;  
  
function setup():void {  
  new Canvas(400, 400);  
  monster = new Sprite(200, 200, 72, 69);  
  monster.image = '👹';  
}  
  
function draw():void {  
  background('grey');  
}
```

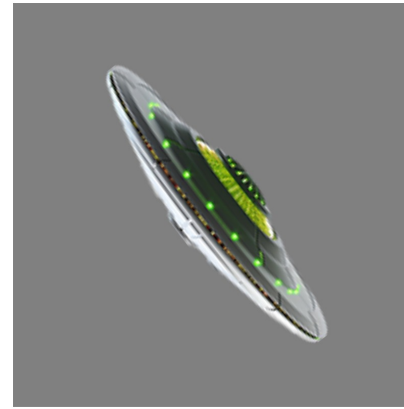


- NOTE: `index.html` must include `<meta charset="utf-8">`

Challenge 5: Sprite Images

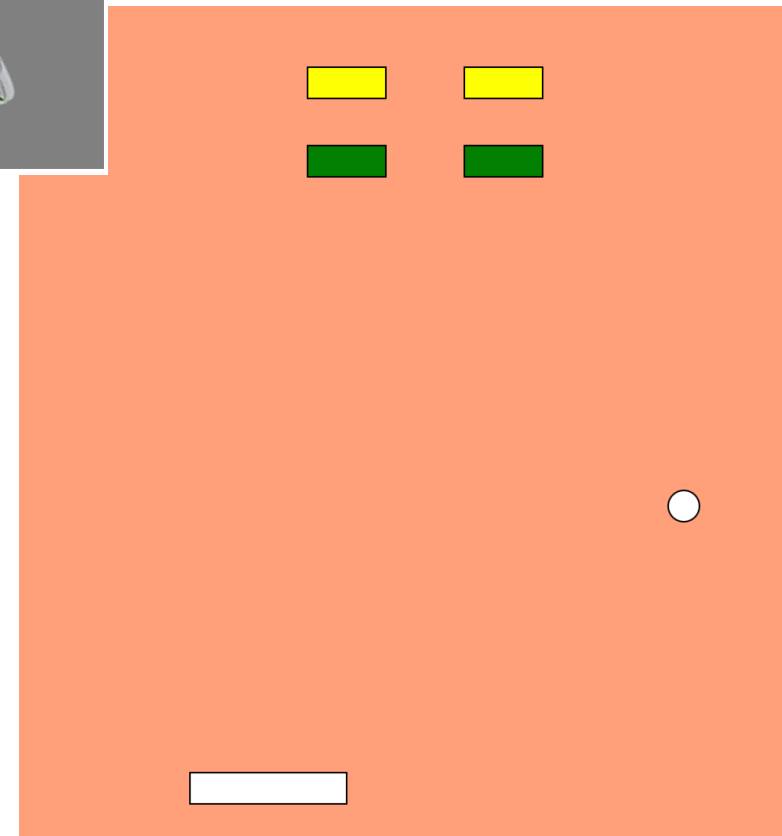
Lesson challenge 5:

- Change monster sprite to a **spaceship**
 - resize to fit the canvas area
 - using the Sprite **scale** property.
 - Perhaps add **rotationSpeed** if you like



Game challenge 5:

- Add some colored blocks sprites at the top of your game screen
- HINT:
 - CTRL-C copy
 - CTRL-V paste
 - Then change **block1** to **block2**, etc.



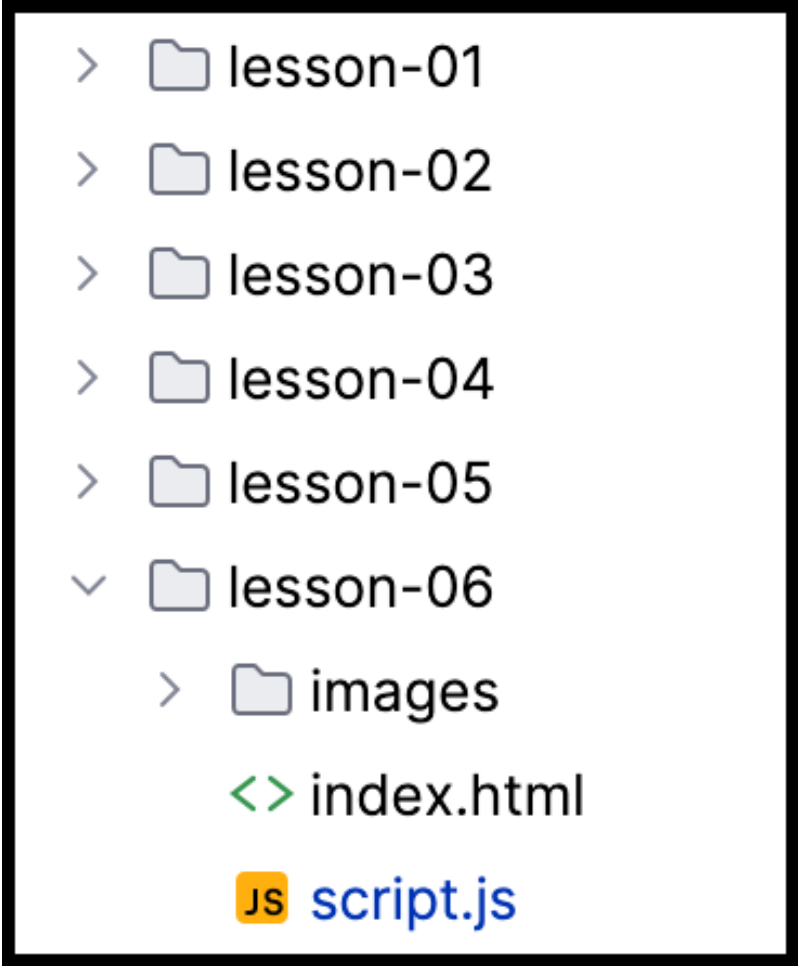
QUESTION

These lessons use PNG images.
Can you name another type of
image format?

Raise your hand if you know the
answer.

Lesson 6: Monster Mover

- Work in folder: **lesson-06**



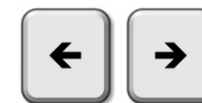
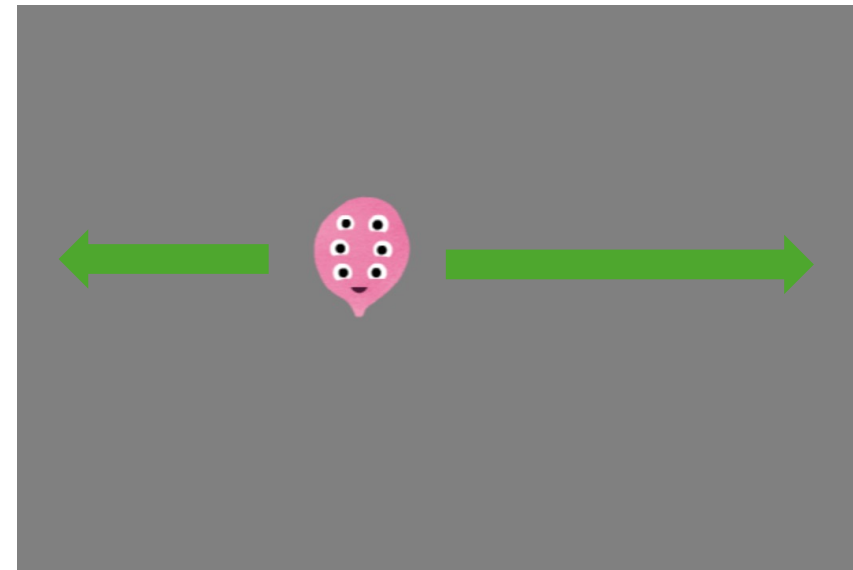
```
> lesson-01
> lesson-02
> lesson-03
> lesson-04
> lesson-05
v lesson-06
  > images
    <> index.html
    JS script.js
```

Lesson 6: Monster Mover

- Our monster wants to move.
- move left and right when we press the left / right arrow keys
- We capture **keypresses** in draw code (with **kb.pressing()**)
 - and change monster's **x** and **y velocity** based on which arrow key was pressed



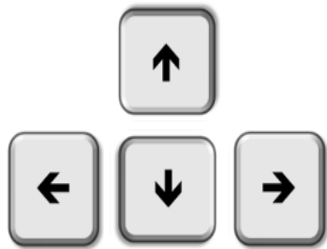
```
function draw() : void {  
  background('grey');  
  
  // reset to zero  
  monster.vel.x = 0;  
  
  if (kb.pressing('left'))  
    monster.vel.x = -5;  
  
  if (kb.pressing('right'))  
    monster.vel.x = 5;  
}
```



Challenge 6: Monster Mover

Lesson challenge 6:

- Add two more **if** tests (for **Y** velocity component)
 - to allow the monster to move up and down as well

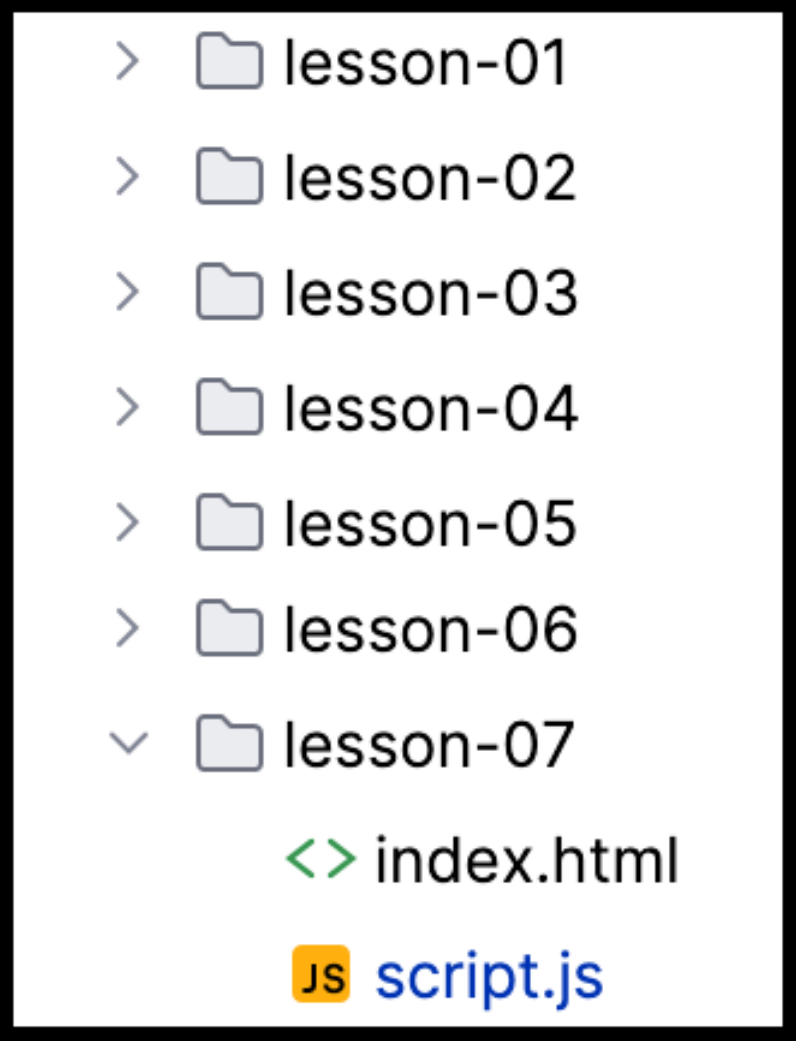


Game challenge 6: let user move paddle left/right

```
function draw() : void {  
    background('grey');  
  
    // reset to zero  
    monster.vel.x = 0;  
    monster.vel.y = 0;  
  
    if (kb.pressing('left'))  
        monster.vel.x = -5;  
  
    if (kb.pressing('right'))  
        monster.vel.x = 5;  
  
    if (kb.pressing('up'))  
        monster.vel.y =           
  
    if (kb.pressing('          '))  
                 = 5;  
}
```

Lesson 7: Gravity & Colliders

- Work in folder: **lesson-07**

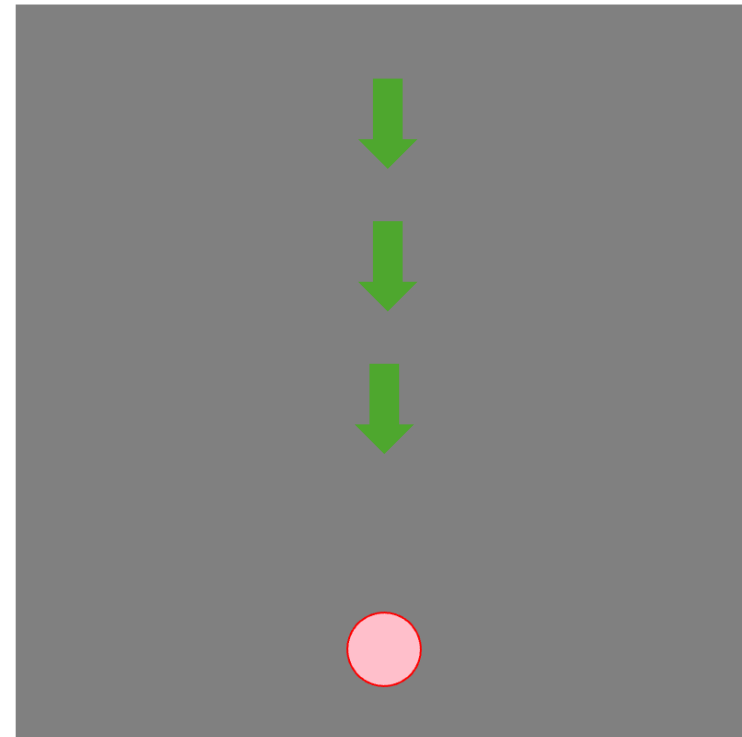


> lesson-01
> lesson-02
> lesson-03
> lesson-04
> lesson-05
> lesson-06
✓ lesson-07
 <> index.html
 JS script.js

Lesson 7: Gravity & Colliders

- By default, our world does not have any gravity. Let's add some gravity to see how it effects a sprite.

```
1 let sprite1;  
2  
3 function setup() : void {  
4   new Canvas(600, 400);  
5  
6   sprite1 = new Sprite(200, 200, 40);  
7   sprite1.colour = 'pink';  
8   sprite1.stroke = 'red';  
9  
10  world.gravity.y = 10;  
11 }  
12  
13 function draw() : void {  
14   background('grey');  
15 }
```



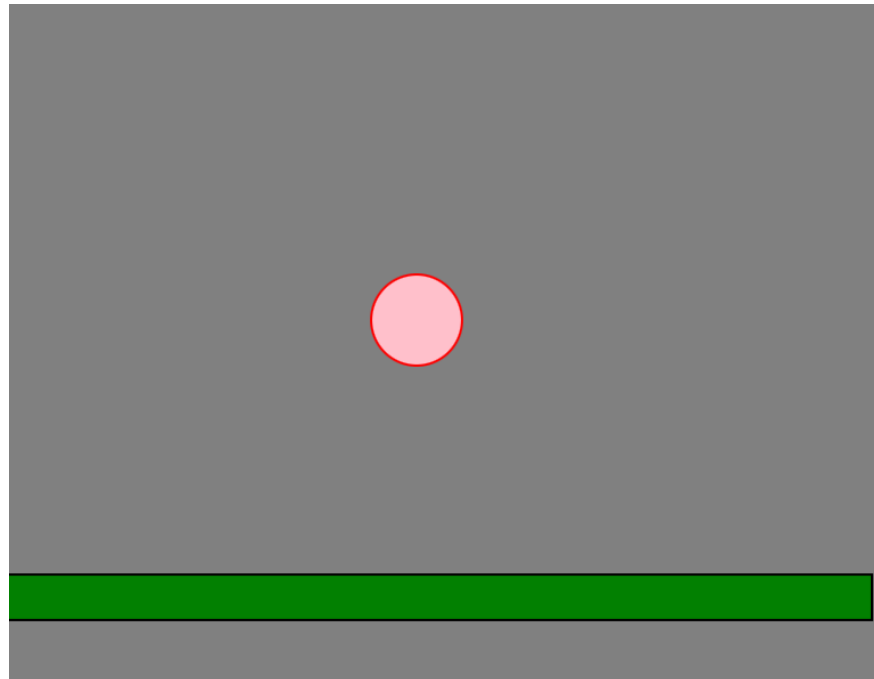
Lesson 7: Gravity & Colliders

- As you can see, the force of gravity causes our sprite to fall downwards and off the canvas. What if we added a floor?

```

1  let sprite, floor;
2
3  function setup() : void {
4    new Canvas(600, 400);
5
6    sprite = new Sprite(200, 200, 40);
7    sprite.colour = 'pink';
8    sprite.stroke = 'red';
9
10   floor = new Sprite(200, 350, 400, 20);
11   floor.colour = 'green';
12   floor.stroke = 'black';
13
14   world.gravity.y = 10;
15 }
16
17 function draw() : void {
18   background('grey');
19 }

```



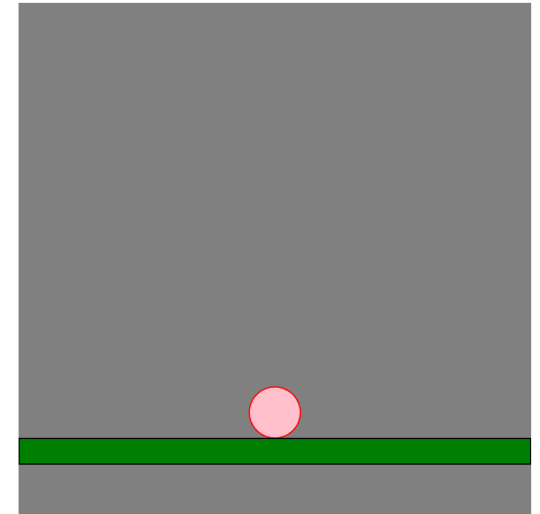
What happens
to the ground
Sprite?

Lesson 7: Gravity & Colliders

- the floor is also affected by gravity and falls off the canvas.
- Default:
 - all sprites have dynamic colliders (affected by gravity & collisions)
- SOLUTION: Declare the floor as STATIC
 - When created

```
floor = new Sprite(200, 350, 400, 20, STATIC);
```
 - Or set physics property

```
floor.physics = STATIC;
```

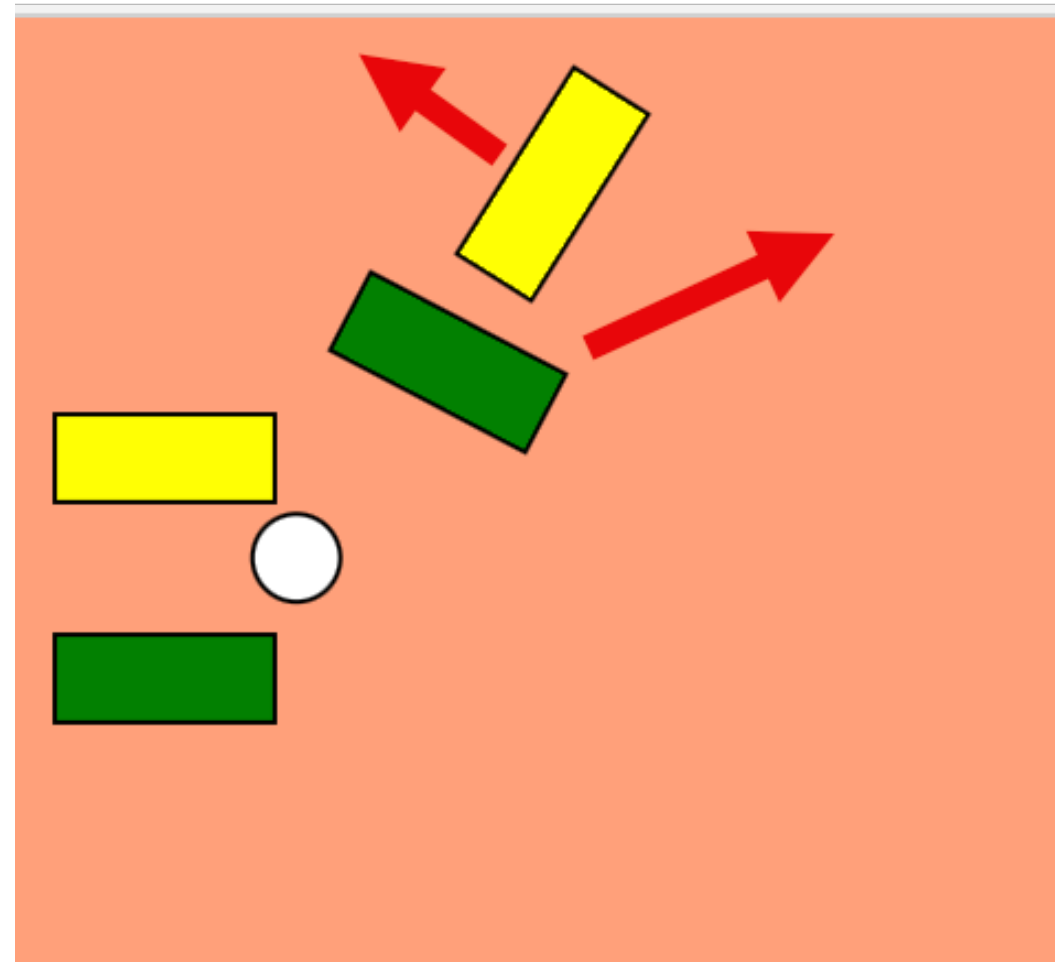


Lesson 7: Gravity & Colliders

Lesson challenge 7:
make the floor STATIC

Game challenge 7:

- Set the paddle to be STATIC
- Set the colored blocks to be STATIC
 - so not moved when hit by ball ...



QUESTION

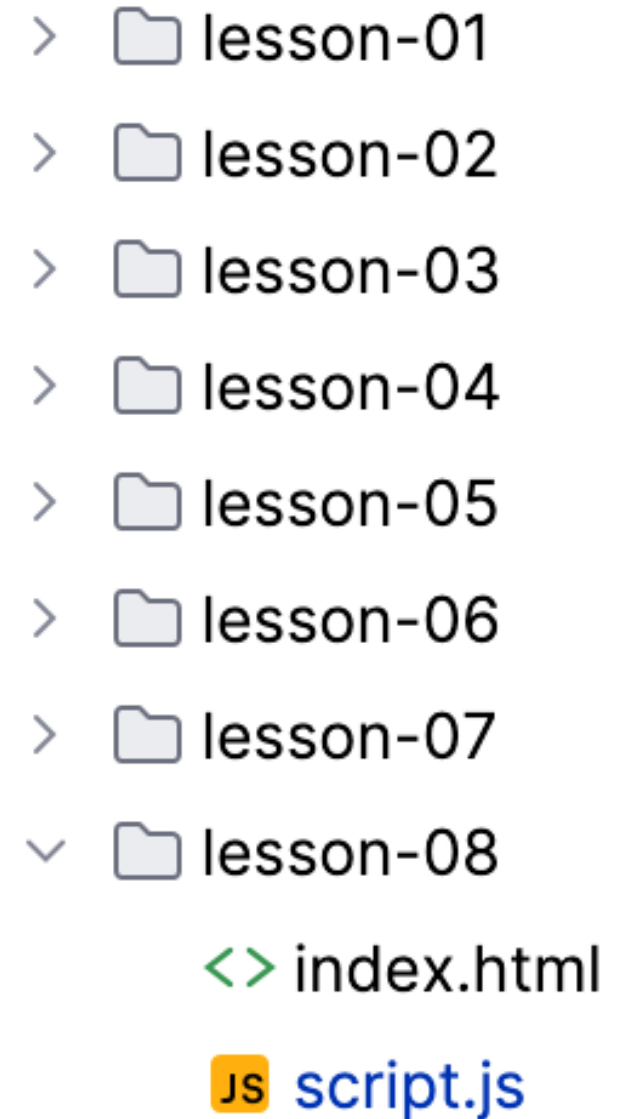
Name the following web browsers?



Raise your hand if you know the answer.

Lesson 8: Screen text

- Work in folder: **lesson-08**



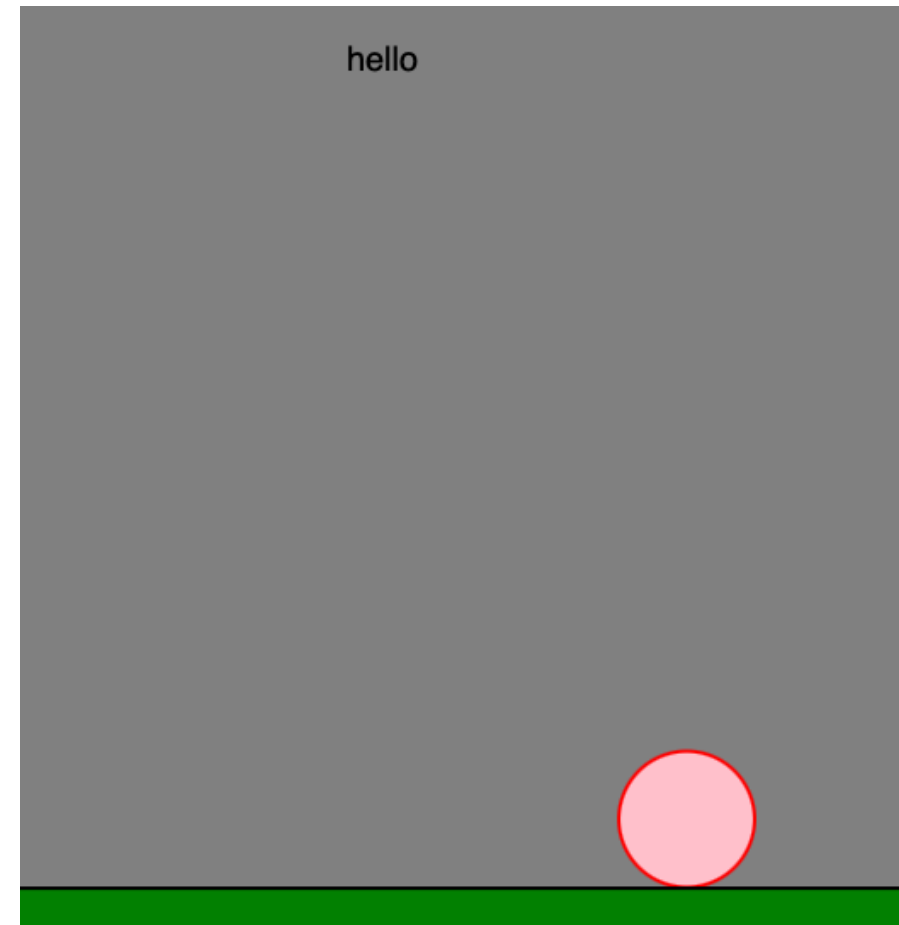
A file explorer view showing a directory structure. It lists eight folders named 'lesson-01' through 'lesson-08'. The 'lesson-08' folder is expanded, showing two files: 'index.html' and 'script.js'. The 'index.html' file is preceded by a green code icon, and 'script.js' is preceded by a yellow 'JS' icon.

- > lesson-01
- > lesson-02
- > lesson-03
- > lesson-04
- > lesson-05
- > lesson-06
- > lesson-07
- ✓ lesson-08
 - <> index.html
 - JS script.js

Lesson 8: Screen text

- It's nice to be able to give information as text
- We can do this with the `text()` command

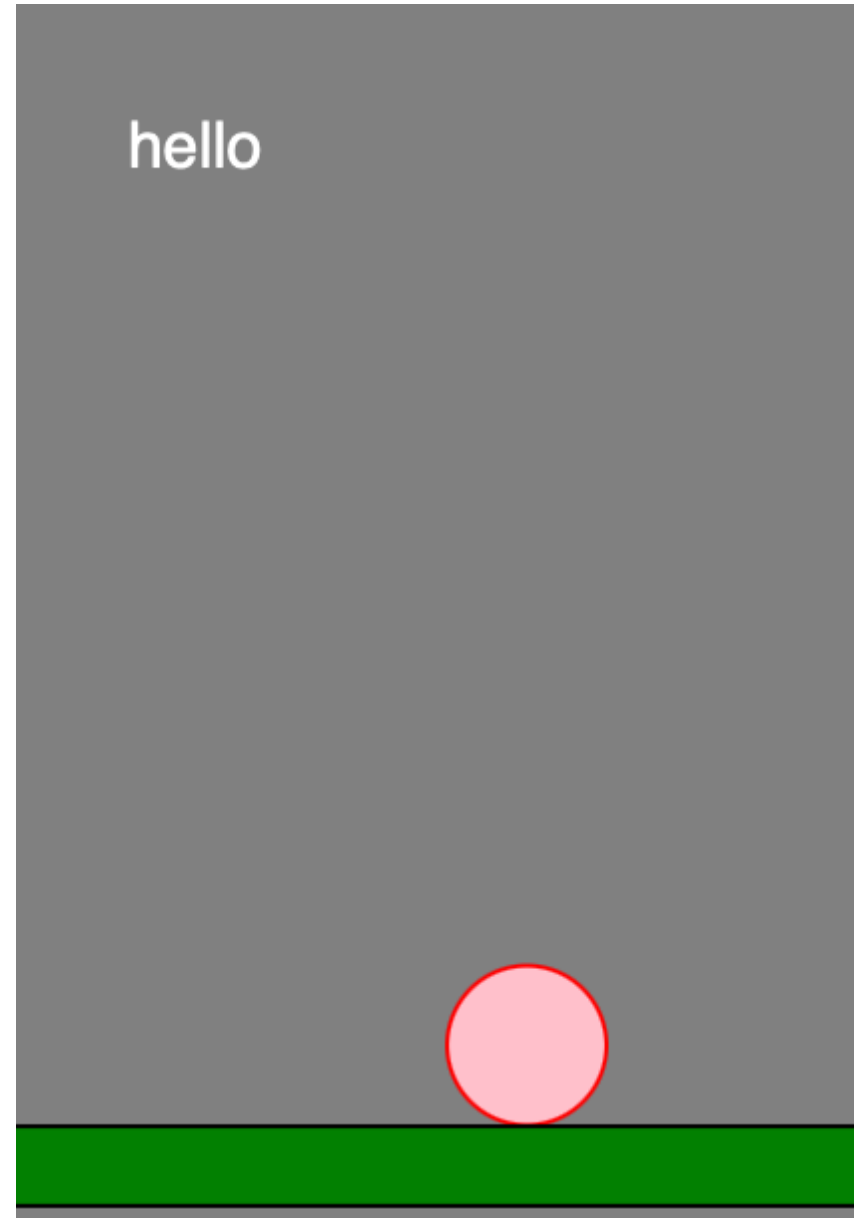
```
function draw() : void {  
  background('grey');  
  
  // text  
  text('hello', 100, 100);  
}
```



Lesson 8: Screen text

- Set text color with command `fill()`
- Set text size with command `textSize()`

```
function draw() : void {  
  background('grey');  
  
  // text  
  fill('white');  
  textSize(16);  
  text('hello', 100, 100);  
}
```



Lesson 8: Screen text

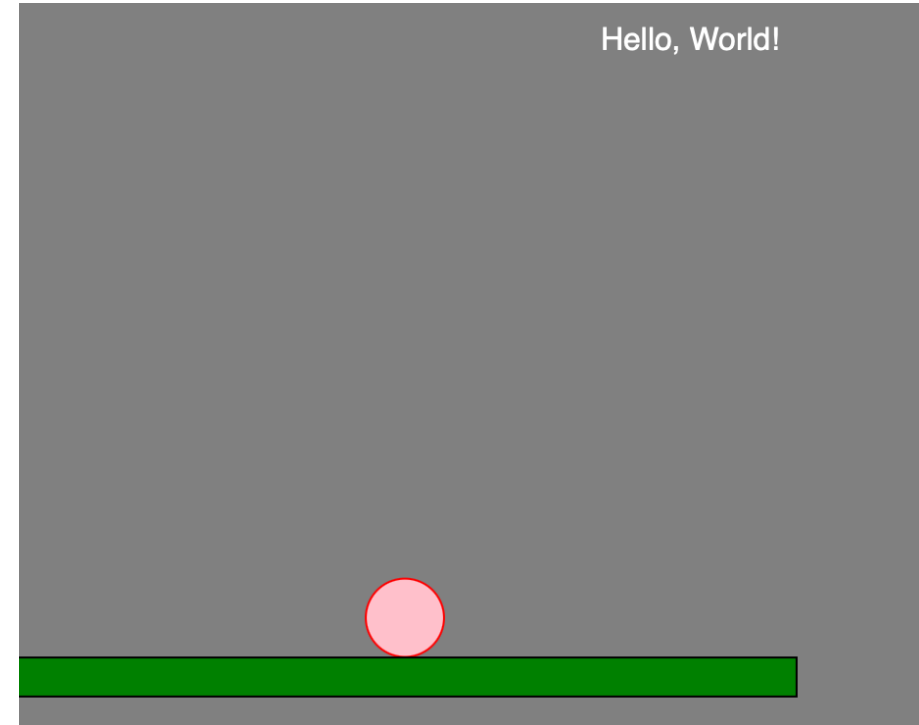
Lesson challenge 8:

write "Hello World!:" In large white text

Game challenge 8:

write " move paddle with arrow keys"

In large white text at the center top of the screen



Lesson 9: Sound

- Work in folder: **lesson-09**

```
> lesson-01
> lesson-02
> lesson-03
> lesson-04
> lesson-05
> lesson-06
> lesson-07
> lesson-08
v lesson-09
    <> index.html
    JS script.js
```

Lesson 9: Sound

- We can preload() sounds, just like images

```
function preload() : void {  
    beepSound = loadSound('sounds/LowVoice.wav');  
}
```

- And they play when with the play() function

```
if(mouse.pressing()) {  
    beepSound.play();  
}
```

Lesson 9: Sound

Lesson challenge 9:

Copy **sounds** folder into your **lesson-09**

Remove the sprites, and display message “click mouse for sound”

Add a variable “beepSound” to the list after keyword “**let**”

Implement the preload() function from the previous slide

In the draw() function, add the code from the previous slide to play a sound when the **mouse** button is **pressing()**

You should hear a beep each time you click a mouse button



click mouse for sound

Lesson 10: Collisions & KINEMATIC Sprites

- Work in folder: **lesson-09**

> lesson-01
> lesson-02
> lesson-03
> lesson-04
> lesson-05
> lesson-06
> lesson-07
> lesson-08
✓ lesson-09

<> index.html
js script.js

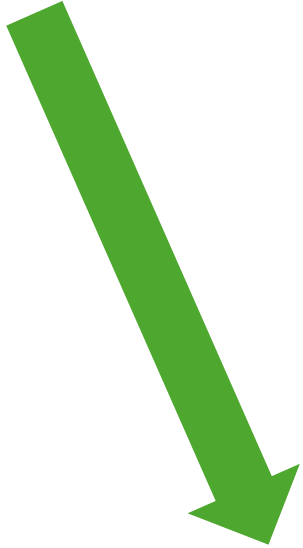
Lesson 10: Collisions & KINEMATIC Sprites

- PROBLEM
 - If our Breakout paddle is STATIC
 - We cannot move it with arrow keys
 - If our Breakout paddle is NOT STATIC
 - It gets moved when hit by ball
- SOLUTION:
 - We declare the paddle as KINEMATIC
 - Meaning we'll move this Sprite through code – not physics of colliding Sprites

Lesson 10: Collisions & KINEMATIC Sprites

- We declare the paddle as KINEMATIC

```
function setup() : void {  
  new Canvas(800, 600);  
  ball = new Sprite(300, 400, 20)  
  ball.color = 'white';  
  
  ball.vel.x = 0.5 * BALL_SPEED;  
  ball.vel.y = BALL_SPEED;  
  
  paddle = new Sprite(200, 550, 100, 20, KINEMATIC);  
  paddle.color = 'white';  
}
```



Lesson 10: Collisions & KINEMATIC Sprites

- PROBLEM
 - Ball just bounces when hits STATIC blocks
 - We want the blocks to be removed
- SOLUTION 1:
 - We **delete()** block when hit
- PROBLEM
 - If we use **delete()** we have errors when testing for collision with block we deleted
- SOLUTION 2:
 - When ball hits block, we MOVE block off screen
 - (more complicated solution to remove deleted block from list of blocks to check ...)

Lesson 10: Collisions & KINEMATIC Sprites

- We detect collision with **overlaps()** function
 - If **ball.overlaps(block<n>)**
 - Set block X to -1000 (off screen)
 - Bounce ball – reverse velocities
- Do this for block1, block2, block3 etc.

```
if (ball.overlaps(block1)) {  
    block1.x = -1000;  
    ball.vel.x = -1 * ball.vel.x;  
    ball.vel.y = -1 * ball.vel.y;  
}
```

```
if (ball.overlaps(block2)) {  
    block2.x = -1000;  
    ball.vel.x = -1 * ball.vel.x;  
    ball.vel.y = -1 * ball.vel.y;  
}
```

```
if (ball.overlaps(block3)) {  
    block3.x = -1000;  
    ball.vel.x = -1 * ball.vel.x;  
    ball.vel.y = -1 * ball.vel.y;  
}
```

Lesson 10: Collisions & KINEMATIC Sprites

Game challenge 10:

Make the paddle KINEMATIC

Test for ball collision with paddle
or any block

- Play a sound on any collision
- Reverse velocity on any collision
- For collision with a block, set the block's X value to -1000
 - So we no longer see it

